

THE UNIVERSITY OF EDINBURGH

School of Mathematics



Strategic Logistics Network Design for Tartan Trade: A
Multi-Period MECWLP Approach

Risk and Logistics
Assignment 1

Prepared By:

1. Selina Liyengwa
2. Samwel Mlabwa
3. Anja Harisoa
4. Mohamed Ali

16 February 2026

1 Introduction

This report presents a strategic logistics network design for Tartan Trade, a start-up company dedicated to the online sale of sustainable Scottish produce under the motto “Think Global! Buy Local!”. The company aims to serve the entirety of Scotland with a commitment to next-day delivery and a minimal carbon footprint achieved through a self-managed supply chain.

The primary objective of this study is to formulate and solve a Multi-Echelon Capacitated Warehouse Location Problem (MECWLP) that determines the optimal locations and timing for warehouse construction over a 10-year planning horizon. Because the company anticipates its customer base growing gradually to a target of 5% of the Scottish population by Year 10, the model focuses on a phased infrastructure build-up to avoid the high costs of underutilized facilities in the early years.

A critical challenge addressed in this report is the inherent uncertainty in long-term demand forecasting. While baseline growth is predicted, the accuracy of these forecasts is assumed to decay by 10% per year, resulting in significant volatility in later periods. To ensure a robust network, the study transitions from a multi-period deterministic model to a stochastic deterministic equivalent formulation using 100 equally likely demand scenarios.

The analysis seeks to minimize the total expected cost, which includes one-off construction setup costs, recurring annual operating costs, and transportation costs from suppliers to warehouses and finally to customers. By utilizing Mixed-Integer Linear Programming (MILP) and data aggregation techniques, this report provides a computationally efficient roadmap for Tartan Trade’s strategic expansion, balancing cost-efficiency with service reliability across a range of possible future demands

2 Data Preparation and Aggregation

The original dataset describes a large-scale multi-period, multi-product, multi-echelon distribution network consisting of 440 customers, 440 candidate warehouse locations, 53 suppliers, 4 product groups, 3 vehicle types, and 10 planning periods. Customer demand is provided at product–period–scenario level, together with geographic coordinates, fixed setup costs, and capacity limits for candidate facilities.

Solving the model, both in deterministic form and in the deterministic equivalent of the stochastic formulation, would generate a prohibitively large number of binary siting and multi-period flow variables. To ensure computational tractability while preserving structural realism, a controlled aggregation and screening procedure was implemented.

2.1 Customer Aggregation via Demand-Weighted K-Means

Customers were aggregated using K-means clustering applied to geographic coordinates. To reflect long-term economic relevance, each customer was weighted by its cumulative 10-year demand across all products, and periods. Let

$$w_i = \sum_{p,t} b_{ipt}$$

denote the total demand of customer i . The clustering minimises the weighted within-cluster sum of squares:

$$\sum_{k=1}^K \sum_{i \in C_k} w_i \| (x_i, y_i) - \mu_k \|^2.$$

Weighting ensures that high-demand districts exert greater influence on centroid location, aligning spatial aggregation with freight intensity rather than purely geographic density. This is consistent with the long-term strategic nature of warehouse siting decisions.

The value $K = 40$ was selected as a practical compromise: smaller values produce geographically large clusters that may poorly represent dispersed customers when intra-cluster transport costs are ignored, while larger values increase model size without significant additional spatial benefit.

Following clustering, demands were aggregated by summation. For the deterministic formulation,

$$b_{kpt} = \sum_{i \in C_k} b_{ipt},$$

and for the stochastic formulation with scenarios s ,

$$b_{kpt}^s = \sum_{i \in C_k} b_{ipt}^s.$$

By construction,

$$\sum_k b_{kpt} = \sum_i b_{ipt}, \quad \sum_k b_{kpt}^s = \sum_i b_{ipt}^s \quad \forall p, t, s,$$

ensuring exact preservation of total system demand under both formulations. Numerical validation confirmed zero aggregation error.

2.2 Candidate Warehouse Reduction via Attractiveness Scoring

Retaining all 440 candidate warehouse locations would result in a large number of binary siting variables. To reduce the decision space while preserving economically attractive options, a structured screening procedure was implemented.

For each cluster k and candidate warehouse j , an attractiveness score $A_{jk} \in [0, 1]$ was computed based on fixed setup cost, proximity to the cluster centroid, and facility capacity. Operating costs were excluded from the score. In the dataset, operating cost is 10% of the fixed setup cost for all candidates and therefore provides no additional discriminatory information. Including both would effectively double-count the cost component, so only fixed setup cost was retained.

Because these criteria are measured in different units, each was transformed using min-max normalisation:

$$X^{norm} = \frac{X - X^{min}}{X^{max} - X^{min}}.$$

Lower fixed cost and shorter distance are preferable and were therefore inverted:

$$\text{score_cost}_j = 1 - F_j^{norm}, \quad \text{score_prox}_{jk} = 1 - d_{jk}^{norm}.$$

Capacity, which is beneficial when larger, was retained directly:

$$\text{score_cap}_j = \text{Cap}_j^{norm}.$$

Distance normalisation was performed separately for each cluster so that candidates were evaluated relative to that cluster's spatial dispersion.

The final attractiveness score was defined as

$$A_{jk} = 0.50 \text{score_prox}_{jk} + 0.25 \text{score_cost}_j + 0.25 \text{score_cap}_j.$$

The weighting reflects the economic structure of facility location: proximity receives the highest weight due to its direct influence on transport cost and service responsiveness, while the remaining weight is equally allocated between fixed cost and capacity to ensure that capital intensity and scalability are also incorporated into the screening process.

For each cluster, candidates were ranked in descending order of A_{jk} and the top 10 were retained. The final reduced candidate set was formed as the union of selected candidates across clusters, with duplicates removed. This screening step does not modify any cost or capacity parameters; it only reduces the feasible siting set prior to optimisation.

2.3 Aggregation Assumptions

The following assumptions were implemented during the data aggregation process:

1. For modeling purposes, all individual customers located within a specific geographic cluster are represented by a single geographic centroid, which serves as the unique point of delivery for that entire region.
2. The total demand assigned to a cluster is calculated as the sum of the demands of all constituent customers within its boundaries.
3. The transportation distance between a warehouse and a customer cluster is approximated by the calculated distance from the facility to the cluster's centroid.
4. Intra-cluster transportation costs (the final leg of delivery from a centroid to an individual customer) are excluded.

3 The MECWLP with Multiple Time Periods

3.1 Mathematical Formulation

Sets and indices

- $i \in I$: set of customers (or aggregated customer groups)
- $j \in J$: set of candidate warehouse locations
- $k \in K$: set of suppliers
- $p \in P$: set of product groups
- $t \in T$: set of time periods

Parameters

- b_{ipt} : demand of customer i for product p in period t
- f_j : setup (construction) cost of warehouse j
- g_j : annual operating cost per unit of capacity at warehouse j
- U_j : capacity of warehouse j
- S_k : annual supply capacity of supplier k
- P_k : product group supplied by supplier k
- c_{ji} : unit transportation cost from warehouse j to customer i
- c_{kj} : unit transportation cost from supplier k to warehouse j

Decision variables

- $y_{jt} \in \{0, 1\}$: is 1 if warehouse j is constructed in period t , 0 otherwise.
- $z_{jt} \in \{0, 1\}$: is 1 if warehouse j is operational in period t , 0 otherwise.
- $x_{jipt} \geq 0$: quantity of product p shipped from warehouse j to customer i in period t
- $w_{kjpt} \geq 0$: quantity of product p shipped from supplier k to warehouse j in period t

Objective function

The objective is to minimize the total cost consisting of warehouse setup costs, operating costs, and transportation costs across all time periods:

$$\min \sum_{j \in J} \sum_{t \in T} f_j y_{jt} + \sum_{j \in J} \sum_{t \in T} g_j z_{jt} + \sum_{j \in J} \sum_{i \in I} \sum_{p \in P} \sum_{t \in T} c_{ji} x_{jipt} + \sum_{k \in K} \sum_{j \in J} \sum_{p \in P} \sum_{t \in T} c_{kj} w_{kjpt} \quad (\text{Objective})$$

Subject to:

$$\sum_{j \in J} x_{jipt} = b_{ipt} \quad \forall i \in I, p \in P, t \in T \quad (1)$$

$$\sum_{i \in I} \sum_{p \in P} x_{jipt} \leq U_j z_{jt} \quad \forall j \in J, t \in T \quad (2)$$

$$x_{jipt} \leq b_{ipt} \cdot z_{jt} \quad \forall i \in I, j \in J, p \in P, t \in T \quad (3)$$

$$\sum_{k \in K} w_{kjpt} = \sum_{i \in I} x_{jipt} \quad \forall j \in J, p \in P, t \in T \quad (4)$$

$$\sum_{j \in J} w_{kjpt} \leq S_k \quad \forall k \in K, p \in P, t \in T \quad (5)$$

$$w_{kjpt} = 0 \quad \forall k \in K, j \in J, t \in T, p \neq P_k \quad (6)$$

$$z_{jt} = \sum_{\tau=1}^t y_{j\tau} \quad \forall j \in J, t \in T \quad (7)$$

$$\sum_{t \in T} y_{jt} \leq 1 \quad \forall j \in J \quad (8)$$

$$y_{jt}, z_{jt} \in \{0, 1\}, x_{jipt}, w_{kjpt} \geq 0 \quad \forall i, j, k, p, t \quad (9)$$

Constraint Explanations

- 1 **Demand satisfaction:** Ensures that for each customer i , product p , and time period t , the total quantity shipped from all warehouses exactly equals the customer's demand b_{ipt} . This is a hard constraint requiring 100% service level in every period.
- 2 **Warehouse capacity:** This constraint limits the total outflow from warehouse j in period t to its maximum capacity U_j , but only if the warehouse is operational ($z_{jt} = 1$). If the warehouse is not operational, no shipments can originate from it. This couples binary operational decisions with continuous flow variables.
- 3 **Disaggregated capacity:** This is a disaggregated constraint that serves two purposes: (i) it forces $x_{jipt} = 0$ whenever warehouse j is not operational ($z_{jt} = 0$), and (ii) it provides a tighter upper bound on individual shipment variables compared to constraint 2 alone, which can improve solver performance.
- 4 **Flow conservation:** Enforces that at each warehouse j , for every product p and period t , the total inbound quantity received from all suppliers must equal the total outbound quantity shipped to all customers. This ensures no product is stored or lost at the warehouse across periods.
- 5 **Supplier capacity:** Limits the total quantity shipped from supplier k in period t to its annual supply capacity S_k . Making sure that suppliers can not ship more than what they have.
- 6 **Product compatibility:** Restricts each supplier k to ship only its designated product group P_k . This encodes the business rule that suppliers are specialized and cannot supply products outside their expertise.
- 7 **Construction-operational link:** This defines the relationship between construction decisions y_{jt} and operational status z_{jt} . A warehouse becomes operational in period t if it was constructed in that period or any previous period. This captures the irreversible nature of construction investments.
- 8 **One-time construction:** Ensures each warehouse can be constructed at most once over the entire planning horizon. This reflects the reality that warehouse construction is a one-time capital investment decision.
- 9 **Variable domains:** Defines the domains of decision variables: binary for construction and operational status, non-negative continuous for shipment quantities.

3.2 Operational Assumptions

To accurately model the strategic expansion of Tartan Trade over its 10-year growth phase, the following Operational Assumptions were implemented to define the scope and mechanics of the multi-period network:

1. For modeling simplicity, we assume that warehouses are constructed instantaneously. A facility commissioned in year t is fully prepared to satisfy customer demand within that same time period
2. Once constructed, a warehouse remains operational until the end of the 10-year planning horizon. The model does not permit the decommissioning of facilities once they have been built.
3. Warehouse closure or capacity adjustment is not permitted. Following initial construction, warehouse closure or capacity adjustments are not permitted. The physical capacity remains constant throughout the facility's lifecycle, with annual operating costs applied consistently until the end of the horizon.
4. A warehouse may serve different customers in different periods as the network grows.
5. A customer may be served by different warehouses in different periods.
6. No inventory is carried between periods. The model assumes that no inventory is carried between periods. Each year's demand realization is treated as a discrete event that must be satisfied entirely by that period's production and transport capacity without reliance on prior stock.

3.3 Cost Assumptions

To accurately reflect the economic impact of the network design over the 10-year horizon, the objective function is governed by the following cost assumptions:

1. The warehouse setup cost is treated as a non-recurring capital expenditure incurred solely in the time period when the facility is first commissioned..
2. Operating cost is incurred in each time period in which the warehouse is operational.
3. Operating cost depends only on installed capacity and not on throughput.
4. Transportation cost is proportional to shipped quantity.
5. To focus on the strategic structural requirements of the network, no financial discounting or inflationary adjustments are applied to future periods; total costs are calculated as the simple arithmetic sum across the 10-year horizon.

3.4 Optimisation Solution Report

This section presents and discusses the results obtained from the proposed multi-period warehouse location model under deterministic demand settings. The model was implemented in FICO Xpress using the Python Programming language and solved as a mixed-integer linear program (MILP). The results identify the optimal warehouse construction and operation schedule, the associated material flows, and the resulting cost structure, including construction, operating, and transportation costs.

Warehouses Construction and Operation Schedule

A total of 13 warehouses are constructed over the 10 years horizon. Table 1 lists the selected warehouses, their construction period, operational periods, capacities, and cost parameters.

Warehouse-Level Performance Over Time

To illustrate the operational dynamics, Table 2 summarises performance in three representative periods (1, 6, and 10), including total flow, available capacity, and utilisation.

Table 1: Warehouse construction and operational decisions

Warehouse	Built in Period	Operational Periods	Capacity (kg)	Setup Cost (£)	Operating Cost (£)
47	2	[2–10]	7,720,000	678,000	67,800
48	1	[1–10]	7,520,000	734,000	73,400
54	1	[1–10]	7,960,000	650,000	65,000
132	6	[6–10]	7,320,000	804,000	80,400
220	6	[6–10]	7,080,000	958,000	95,800
282	1	[1–10]	7,280,000	832,000	83,200
315	1	[1–10]	7,120,000	930,000	93,000
339	1	[1–10]	7,480,000	734,000	73,400
344	1	[1–10]	7,400,000	776,000	77,600
350	1	[1–10]	7,600,000	706,000	70,600
398	1	[1–10]	7,160,000	902,000	90,200
413	1	[1–10]	7,480,000	748,000	74,800
439	6	[6–10]	7,000,000	1,000,000	100,000

Table 2: Network performance summary by period (selected periods)

Period	Operational Warehouses	Total Inbound (kg)	Total Capacity (kg)	Overall Utilisation (%)
1	9	4,317,994	67,000,000	6.4
6	13	25,922,970	96,120,000	27.0
10	13	43,058,834	96,120,000	44.8

Cost Summary

The optimization model returns a feasible warehouse network plan with a total cost:

$$\textbf{Total Cost} = \text{£}38,050,043.05.$$

The available cost components are:

$$\text{Setup (Construction) Costs} = \text{£}10,452,000.00$$

$$\text{Operating Costs} = \text{£}9,003,200.00$$

$$\text{Transportation costs} = \text{£}18,594,843.05$$

Runtime and Solution Quality

The model was solved using FICO Xpress with a maximum time limit of 15 minutes. The solver explored 245 branch-and-bound nodes within this time and identified two feasible integer solutions. The best feasible solution obtained has a total cost of:

$$\textbf{Final MIP Objective} = \text{£}38,050,043.05.$$

However, global optimality was not proven before the time limit was reached. The final best bound reported by the solver is:

$$\textbf{Final MIP Bound} = \text{£}22,930,860.61,$$

resulting in a **MIP gap of 39.73%**.

CO2 Emission

The deterministic network design results in total customer-leg transport emissions of approximately **4.93 million kgCO₂** over the planning horizon. This reflects the environmental impact of the fixed demand scenario and highlights the carbon intensity associated with the selected warehouse locations and distribution flows.

3.5 Results Interpretation

The optimization model delivers a feasible warehouse network configuration with a total projected cost of £38.05M over the planning horizon. While the solution was obtained within a limited computational time frame and is not formally proven optimal (remaining MIP gap of 39.73%), it represents a high-quality, implementable design based on rigorous mathematical optimization.

The selected configuration prioritizes early construction of large-capacity facilities, indicating that economies of scale and long-term operational stability are the dominant cost drivers. The resulting network is strategically centralized and capacity-efficient, reducing the need for staggered infrastructure expansion in later periods.

However, given the remaining optimality gap, the reported cost should be viewed as a conservative estimate. Additional computational time could potentially identify lower-cost refinements, particularly in marginal location or timing adjustments. From a strategic standpoint, the current solution provides a robust baseline scenario suitable for investment planning, subject to further sensitivity testing on demand forecasts and cost assumptions.

Overall, the model offers a strong decision-support foundation, enabling informed capital allocation while clearly identifying areas where additional optimisation effort may yield further financial improvements.

3.6 Sensitivity Analysis: Impact of Transportation Cost Reduction

To evaluate the sensitivity of the proposed multi-period warehouse location model to changes in transportation costs, a parametric analysis was conducted by uniformly reducing the vehicle cost per mile per tonne by **£0.001** across all vehicle types. This modification represents improved transportation efficiency or access to a lower-cost logistics service provider, while keeping all other model parameters unchanged. The resulting solution is compared against the baseline model in terms of total cost, warehouse configuration, capacity utilization, and computational performance.

Table 3: Comparison of baseline and sensitivity models

Metric	Baseline Model	Sensitivity Model
Total cost (£million)	38.05	35.27
Cost change (%)	–	-7.3
Warehouses opened	13	11
Overall utilization at period 10 (%)	44.8	52.7
MIP gap (%)	39.73	34.09

As shown in Table 3, the sensitivity model yields a lower total system cost compared to the baseline solution, corresponding to a **7.3%** reduction (approximately **£2.78 million** in total system cost). This cost improvement is accompanied by a more compact network structure, with 2 fewer warehouses opened, and a higher overall utilization of warehouse capacity (from **44.8 %** to **52.7%**).

From a computational standpoint, the modified instance remains challenging, reaching the imposed time limit with a final MIP gap of **34.09%**, compared to **39.73%** for the baseline case. The improved best feasible solution and tighter optimality gap suggest that lower transportation costs lead to a smoother cost landscape, facilitating improved solution quality within the same computational budget. These results suggest that the baseline solution is not only robust to parameter changes but can also be improved through targeted parameter adjustments.

4 The MECWLP with multiple time periods and uncertain demands

4.1 Mathematical Formulation

Sets and indices

- $i \in I$: set of customers
- $j \in J$: set of candidate warehouse locations
- $k \in K$: set of suppliers
- $p \in P$: set of product groups
- $t \in T$: set of time periods
- $s \in S$: set of Scenarios

Parameters

- b_{ipt} : predicted demand of customer i for product p in period t
- $\hat{b}_{ipt}^s$: realized demand of customer i for product p in period t under scenario s
- π_s : probability of scenario s , with $\pi_s = \frac{1}{|S|} = 0.01$
- f_j : setup (construction) cost of warehouse j
- g_j : annual operating cost per unit of capacity at warehouse j
- U_j : capacity of warehouse j
- S_k : annual supply capacity of supplier k
- P_k : product group supplied by supplier k
- c_{ji} : unit transportation cost from warehouse j to customer i
- c_{kj} : unit transportation cost from supplier k to warehouse j

The realized demand satisfies:

$$\hat{b}_{ipt}^s \in [(1 - 0.1t)b_{ipt}, (1 + 0.1t)b_{ipt}] \quad \forall i, p, t, s$$

Decision variables

First-stage Decisions: These decisions must be made before uncertainty is revealed and cannot be changed between scenarios.

- $y_{jt} \in 0, 1$: equals 1 if warehouse j is constructed in period t , 0 otherwise.
- $z_{jt} \in 0, 1$: equals 1 if warehouse j is operational in period t , 0 otherwise.

Second-stage Decisions (scenario-dependent): Recourse actions taken after the specific demand realization for scenario s is known.

- $x_{jip}^s \geq 0$: quantity of product p shipped from warehouse j to customer i in period t under scenario s
- $w_{kjp}^s \geq 0$: quantity of product p shipped from supplier k to warehouse j in period t under scenario s

Objective function

The objective is to minimize the expected total cost consisting of warehouse setup costs, operating costs, and expected transportation costs across all time periods and scenarios.

$$\min \sum_{j \in J} \sum_{t \in T} f_j y_{jt} + \sum_{j \in J} \sum_{t \in T} g_j z_{jt} + \sum_{s \in S} \pi_s \left[\sum_{j \in J} \sum_{i \in I} \sum_{p \in P} \sum_{t \in T} c_{ji} x_{jipt}^s + \sum_{k \in K} \sum_{j \in J} \sum_{p \in P} \sum_{t \in T} c_{kj} w_{kjpt}^s \right] \quad (\text{Objective})$$

Subject to:

$$\sum_{j \in J} x_{jipt}^s = \hat{b}_{ipt}^s \quad \forall i \in I, p \in P, t \in T, s \in S \quad (1)$$

$$\sum_{i \in I} \sum_{p \in P} x_{jipt}^s \leq U_j z_{jt} \quad \forall j \in J, t \in T, s \in S \quad (2)$$

$$x_{jipt}^s \leq \hat{b}_{ipt}^s \cdot z_{jt} \quad \forall i \in I, j \in J, p \in P, t \in T, s \in S \quad (3)$$

$$\sum_{k \in K} w_{kjpt}^s = \sum_{i \in I} x_{jipt}^s \quad \forall j \in J, p \in P, t \in T, s \in S \quad (4)$$

$$\sum_{j \in J} w_{kjpt}^s \leq S_k \quad \forall k \in K, p \in P, t \in T, s \in S \quad (5)$$

$$w_{kjpt}^s = 0 \quad \forall k \in K, j \in J, t \in T, p \neq P_k, s \in S \quad (6)$$

$$z_{jt} = \sum_{\tau=1}^t y_{j\tau} \quad \forall j \in J, t \in T \quad (7)$$

$$\sum_{t \in T} y_{jt} \leq 1 \quad \forall j \in J \quad (8)$$

$$y_{jt}, z_{jt} \in \{0, 1\}, x_{jipt}^s, w_{kjpt}^s \geq 0 \quad \forall i, j, k, p, t, s \quad (9)$$

Constraint Explanations

- 1 **Stochastic demand satisfaction:** For each scenario s , ensures that the total quantity shipped to customer i for product p in period t exactly equals the realized demand $\hat{b}_{ipt}^s$ under that scenario. This must hold for every possible future demand realization, making this a set of $|S|$ parallel constraints.
- 2 **Warehouse capacity under uncertainty:** Under each scenario s , limits the total outflow from warehouse j in period t to its physical capacity U_j , but only if the warehouse is operational. Note that the operational status z_{jt} is scenario-independent (first-stage), while the flow variables x_{jipt}^s can adapt to each scenario. This creates the core two-stage structure: capacity decisions are fixed before uncertainty is revealed, but flow decisions can react.
- 3 **Disaggregated capacity under uncertainty:** This is a disaggregated constraint that links first-stage operational decisions z_{jt} to second-stage shipment variables x_{jipt}^s . It prevents shipments from inactive warehouses while also providing tighter bounds that improve computational performance.
- 4 **Flow conservation under uncertainty:** For each scenario s , enforces that at every warehouse j , product p , and period t , the total inbound quantity from suppliers exactly matches the total outbound quantity to customers. This must hold independently for each demand realization.
- 5 **Supplier capacity under uncertainty:** Under each scenario s , limits supplier shipments to their annual capacity S_k . Since supplier capacities are fixed and cannot be expanded reactively, this constraint must hold for every possible demand scenario.

- 6 **Product compatibility under uncertainty:** Enforces that supplier k can only ship its designated product group P_k under all scenarios. This specialization is a structural business rule that cannot be violated regardless of demand realization.
- 7 **Construction-operational link:** This is a first-stage constraint defining the relationship between construction and operational decisions. Since these decisions are made before uncertainty is revealed, they are identical across all scenarios.
- 8 **One time construction:** Ensures each warehouse can be constructed at most once across all time periods. It reflects the irreversible nature of facility investment decisions.
- 9 **Variable domains:** Defines the domains of all decision variables: binary first-stage variables (y_{jt}, z_{jt}) that are scenario-independent, and non-negative continuous second-stage variables $(x_{j|pt}^s, w_{k|pt}^s)$ that can adapt to each scenario s .

4.2 Two-Stage Stochastic Model Results

This section presents the computational results obtained from the two-stage stochastic warehouse location model. The model explicitly captures demand uncertainty through a finite set of equally likely scenarios and determines optimal long-term warehouse location decisions together with scenario-dependent operational flows. The computational study considers 40 customer districts, 106 candidate warehouse locations, and 53 suppliers. Demand is defined over 4 product groups and 10 time periods, with uncertainty represented by 20 equally likely scenarios.

Warehouse Construction and Operation Decisions

The model recommends constructing 12 warehouses over the planning horizon. Construction is phased strategically across periods to hedge against demand uncertainty while avoiding excessive early fixed-cost commitments, as summarized in the table 4 below.

Table 4: Warehouse construction and operational decisions (Stochastic Model)

Warehouse	Built in Period	Operational Periods	Capacity (kg)	Setup Cost (£)	Operating Cost (£)
19	6	[6–10]	7,880,000	650,000	65,000
47	7	[7–10]	7,720,000	678,000	67,800
48	1	[1–10]	7,520,000	734,000	73,400
54	4	[4–10]	7,960,000	650,000	65,000
58	9	[9–10]	7,960,000	650,000	65,000
284	1	[1–10]	7,880,000	650,000	65,000
345	8	[8–10]	8,000,000	650,000	65,000
351	7	[7–10]	7,920,000	650,000	65,000
363	9	[9–10]	7,840,000	664,000	66,400
401	1	[1–10]	7,800,000	664,000	66,400
411	1	[1–10]	7,960,000	650,000	65,000
423	1	[1–10]	7,960,000	650,000	65,000

Warehouse Performance Over Time

To illustrate the operational evolution over time, Table 5 summarizes performance in three representative periods (1, 6, and 10), including total capacity, and percentage utilization.

Table 5: Network performance summary by period (selected periods)

Period	Operational Warehouses	Total Capacity (kg)	Expected Overall Utilization (%)
1	5	39,120,000	2.2
6	7	54,960,000	9.1
10	12	94,400,000	8.6

Objective Value and Cost Breakdown

The total expected cost over the planning horizon is:

$$\textbf{£17,542,208.82}$$

The deterministic (first-stage) cost components are:

$$\text{Setup (Construction) Costs} = \text{£7,940,000.00}$$

$$\text{Operating Costs} = \text{£5,117,000.00}$$

$$\text{Transportation Costs} = \text{£4,485,208.82}$$

Runtime and Solution Quality

The stochastic model was solved using FICO Xpress with a maximum time limit of 15 minutes (900 seconds). Within this time, the solver primarily operated at the root node and identified one feasible integer solution through heuristic procedures. The best feasible solution obtained has a total expected cost of:

$$\textbf{Final MIP Objective} = \textbf{£17,542,208.82}$$

However, global optimality was not proven before the time limit was reached. The final best bound reported by the solver is:

$$\textbf{Final MIP Bound} = \textbf{£14,315,486.52}$$

resulting in a MIP gap of:

$$\textbf{MIP Gap} = \textbf{18.39\%}$$

CO₂ Emission Results

The stochastic network configuration results in total expected transport emissions of approximately **1.17 million kgCO₂** over the planning horizon. This value is scenario-weighted and accounts for round-trip distances, realized second-stage flows, and vehicle-specific emission factors. It provides a quantitative sustainability indicator that complements the cost analysis, highlighting the environmental impact of warehouse location and flow allocation decisions under demand uncertainty.

4.3 Results Interpretation

The optimization model delivers a feasible stochastic warehouse network configuration with a total expected cost of **£17.54M** over the planning horizon. While the solution was obtained within a limited computational time frame and is not formally proven optimal (remaining MIP gap of **18.39%**), it represents a high-quality, implementable design derived from rigorous two-stage stochastic optimization.

The selected configuration reflects strategic timing of warehouse construction across the horizon, balancing early infrastructure investment with later-stage capacity expansion in response to uncertainty. The solution demonstrates structured capacity allocation across periods, indicating that the model effectively internalizes demand variability across the 20 scenarios considered.

However, given the remaining optimality gap, the reported cost should be interpreted as an upper bound on the true optimal expected cost. Additional computational time could potentially strengthen the bound and identify lower-cost refinements, particularly in marginal construction timing or network configuration adjustments.

From a strategic standpoint, the current solution provides a robust baseline scenario for capital planning under demand uncertainty. Nevertheless, further analysis—such as increasing the number of demand scenarios, performing sensitivity testing on cost parameters, or improving numerical scaling—could enhance solution quality and strengthen optimality guarantees.

Overall, the stochastic model provides a strong decision-support framework, enabling informed long-term infrastructure planning while explicitly accounting for demand uncertainty and highlighting areas where additional optimization effort may yield further financial improvements.

5 Less Conservative Network Formulations

The case study data shows that demand uncertainty grows by 10% annually, reaching a potential 200% of the forecast by Year 10. In a conservative model, these rare, high-demand outliers exert an unduly large influence on the solution, forcing the construction of warehouses that remain underutilized in the vast majority of scenarios. To optimize Tartan Trade’s strategic build-up, two less conservative approaches are proposed.

5.1 Unmet Demand Formulation

This approach treats customer demand as a soft constraint. Instead of committing to permanent infrastructure to handle every possible peak, the model is permitted to leave a portion of demand unsatisfied if the cost of doing so is lower than the capital expenditure.

Define a new non-negative recourse variable $u_{ipt}^s \geq 0$ representing the quantity of demand for customer i , product group p , in period t and scenario s that remains unsatisfied. Let M be a high penalty cost per unit of unmet demand. The modified objective function will now be as indicated in equation 1 below. On the other hand, the demand satisfaction constraint will also be modified to account for unsatisfied demand u_{ipt}^s in each scenario as shown in equation 2 below.

Modified Objective Function:

$$\min \sum_{j \in J} \sum_{t \in T} (f_j y_{jt} + g_j z_{jt}) + \sum_{s \in S} \pi_s \left[\sum_{j,i,p,t} c_{ji} x_{jip}^s + \sum_{k,j,p,t} c_{kj} w_{kjp}^s + \sum_{i,p,t} M \cdot u_{ipt}^s \right] \quad (1)$$

Modified Demand Satisfaction Constraint:

$$\sum_{j \in J} x_{jip}^s + u_{ipt}^s = \hat{b}_{ipts}^s \quad \forall i \in I, p \in P, t \in T, s \in S \quad (2)$$

Large M : A constant larger than any possible demand sum.

This allows the solver to perform a cost-benefit analysis. If the strategic cost of building a warehouse is higher than the expected penalty of not serving rare high-demand peaks, the model will prioritize cost-efficiency over 100% service levels.

5.2 Penalty for Capacity Violation (Overtime/Overflow)

This strategy recognizes that physical capacity is often flexible at a price. Rather than building oversized permanent warehouses, the system can temporarily exceed rated capacity U_j by utilizing overtime labor or external third-party storage during peak realizations. This strategy assumes that a physical warehouse capacity can be temporarily exceeded through overtime labor or external storage at a variable cost (V).

Capacity Violation Formulation

In this case, we define a new non-negative recourse variable $o_{jt}^s \geq 0$ representing the overflow quantity handled by warehouse j in period t under scenario s . Let V be the premium operating cost per unit of overflow. Therefore we allow the violation of the capacity constraint and add the associated cost in the objective function as shown in 3. The new capacity warehouse will also be modified as indicated in 4 below.

Modified Objective Function:

$$\min \sum_{j \in J} \sum_{t \in T} (f_j y_{jt} + g_j z_{jt}) + \sum_{s \in S} \pi_s \left[\sum_{j,i,p,t} c_{ji} x_{jipt}^s + \sum_{k,j,p,t} c_{kj} w_{kjpg}^s + \sum_{j,t} V \cdot o_{jt}^s \right] \quad (3)$$

Modified Capacity Constraint:

$$\sum_{i \in I} \sum_{p \in P} x_{jipt}^s \leq U_j z_{jt} + o_{jt}^s \quad \forall j \in J, t \in T, s \in S \quad (4)$$

Physical capacity is often flexible at a price. This formulation prevents a single unlikely high-demand scenario from forcing the permanent construction of an oversized warehouse that remains underutilized most of the time.

Combining the 2 approaches in the model with the respective modified demand satisfaction and capacity constraints, the objective function is:

Unified Objective Function:

$$\min \sum_{j \in J} \sum_{t \in T} (f_j y_{jt} + g_j z_{jt}) + \sum_{s \in S} \pi_s \left[\sum_{j,i,p,t} c_{ji} x_{jipt}^s + \sum_{k,j,p,t} c_{kj} w_{kjpg}^s + \sum_{i,p,t} M \cdot u_{ipt}^s + \sum_{j,t} V \cdot o_{jt}^s \right] \quad (5)$$

With this formulation, the solver performs a continuous cost-benefit analysis between fixed strategic costs (f_j, g_j) and the expected cost of penalties across all 100 scenarios. If the cost of building a permanent warehouse to handle a rare Year 10 spike in demand is higher than the expected penalty of paying M or V for that rare event, the model will choose not to build.

For the combined formulation to behave realistically/rationally, the penalties must be calibrated relative to each other and the transportation costs. V (Overflow Cost) should be set as a premium (e.g., 50-100%) above standard operating costs but lower than M . This ensures the solver tries to meet demand by paying for overtime/overflow before it considers not meeting the demand entirely. M should be very high, representing the lost goodwill or gross margin of the product. If the penalties are too low, the model will pay penalties instead of building warehouses even during normal demand years. M and V should be high enough that they only trigger during the most extreme high-demand scenarios, preventing those outliers from having an unduly large influence on the 10-year build schedule.

6 Discussions and Recommendations

6.1 Comparative Analysis of Deterministic and Stochastic Models

The deterministic and stochastic formulations yield markedly different network designs, highlighting the substantial impact of demand uncertainty on strategic warehouse location decisions. Table 6 summarizes the key differences between the two solutions.

Table 6: Comparison of Deterministic and Stochastic Model Solutions

Metric	Deterministic Model	Stochastic Model
Number of warehouses opened	13	12
Total cost (£million)	38.05	17.54
Setup costs (£million)	10.45	7.94
Operating costs (£million)	9.00	5.12
Transportation costs (£million)	18.59	4.49
MIP gap (%)	39.73	18.39

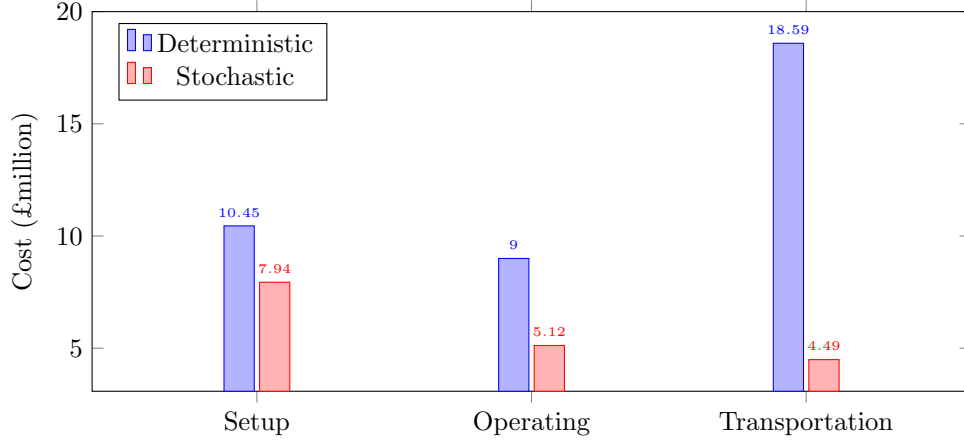


Figure 1: Cost component comparison between deterministic and stochastic models

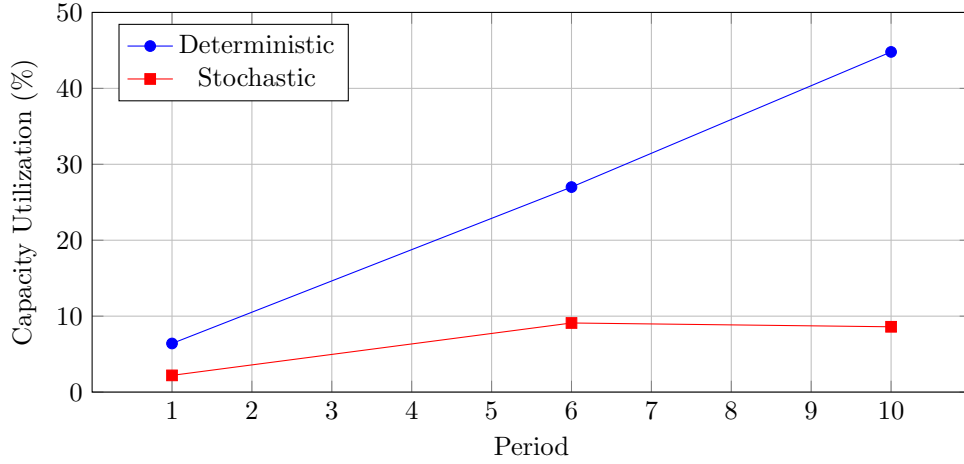


Figure 2: Capacity utilization comparison over selected periods

Key Discussion Points

The comparative analysis reveals several important insights:

- **Network Size and Structure:** The deterministic model opens 13 warehouses, while the stochastic model opens only 12. This reduction reflects the stochastic model's reluctance to commit capital to facilities that may prove unnecessary under many demand scenarios.

- **Cost Structure:** The most striking difference is in transportation costs (£18.59M deterministic vs £4.49M stochastic). The deterministic model, assuming perfect foresight, builds larger facilities and locates them to minimize transport costs. The stochastic model, facing uncertainty, adopts a more conservative posture with lower fixed commitments and accepts higher expected recourse costs.
- **Construction Timing:** The deterministic model heavily front-loads construction (10 of 13 warehouses built in period 1), while the stochastic model staggers construction across periods (only 5 built in period 1). This phasing provides flexibility to adjust capacity as demand uncertainty resolves over time.
- **Capacity Utilization:** The deterministic model achieves steadily increasing utilization, reaching 44.8% by period 10. The stochastic model maintains lower utilization (8.6% by period 10), indicating deliberate overcapacity to hedge against upside demand scenarios.

6.2 Recommendations

Based on the comparative analysis, we recommend the following:

- **Adopt the stochastic framework:** The stochastic model’s phased construction (only 5 of 12 warehouses built in period 1) provides flexibility that the deterministic model’s front-loaded approach lacks. This delayed commitment preserves option value as demand uncertainty resolves over time.
- **Implement flexible capacity mechanisms:** Given the extreme demand uncertainty in later periods (up to 100% by Year 10), the less conservative formulations in Section 5 (unmet demand penalties or temporary capacity overflow) should be explored to avoid over-investing for rare high-demand scenarios.
- **Improve solution quality:** The 18.39% optimality gap can be reduced by extending computational time beyond 15 minutes or exploring decomposition methods tailored for two-stage stochastic programs.

6.3 Conclusion

This report outlined formulation and solving both deterministic and stochastic multi-period MECWLP models for Tartan Trade’s Scottish distribution network. The stochastic solution (12 warehouses, £17.54M expected cost) offers a more resilient strategy than the deterministic alternative, explicitly accounting for demand uncertainty through scenario-based planning. While computational limits prevent optimality guarantees, the solutions provide actionable insights for phased infrastructure investment. The proposed less conservative formulations offer pathways to further balance capital commitment against service reliability, supporting Tartan Trade’s goal of a cost-effective, sustainable supply chain aligned with its long-term growth objectives.